

# Device Tracking Error and Headset Slip: Two Error Sources in HMD-Reported Head Pose for Rehabilitation

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## Abstract

When a clinician uses a Virtual Reality (VR) head-mounted display (HMD) to measure the position, angle or other derived biomechanical quantities, like range of motion (ROM), this measurement can accumulate errors due to two limitations: the ability of the device to record its own pose accurately, and the relative displacement of the HMD against the user's head. We have measured both error sources on a Meta Quest 3S headset worn by a person against an optical motion capture (OMC) system, that was considered the source of truth. Preliminary results indicate that the tracking accuracy of a worn headset is comparable to values previously reported for HMDs rigidly mounted onto tripods or mannequin heads. Under the tested conditions, headset slip relative to the head stayed below the device tracking error but varied by more than an order of magnitude depending how well the headset was fitted.

## Introduction

The use of head-mounted displays (HMDs) in rehabilitation has gradually evolved with time. First used to deliver engaging therapeutic content, they are now measuring instruments in their own right: the pose the HMD reports can be recorded and analyzed to assess the patient's progress (Santos-Paz et al. 2022; Forsberg et al. 2023). Cervical range of motion (ROM), one of the most common assessments in neck rehabilitation, is usually measured with a universal goniometer: small, portable and low-cost, but dependent on an experienced practitioner for a reliable reading (Santos-Paz et al. 2022). The alternative, an optical motion capture laboratory, is accurate but costly, fixed in place and available to very few clinics. A consumer HMD sits between the two, offering instrumented, operator-independent measurement at a cost an

ordinary hospital (or even a patient) can absorb; recent work reports excellent reliability for VR-based cervical assessment, with no significant difference between high-end and low-cost standalone headsets (Santos-Paz et al. 2026), and high concurrent validity against a gold-standard optical system (Forsberg et al. 2023).

Multiple studies have evaluated HMD tracking accuracy, reporting errors on the order of millimeters (Niehorster et al. 2017; Banaszczyk et al. 2024). In these evaluations the HMD is held rigidly in place on a tripod or a mannequin head, to isolate the device from any other source of error. For a clinical purpose these measurements are therefore incomplete: slippage caused by head movements, facial expressions, speech or a loosening headstrap are not accounted for.

The stability of the mounting itself has been examined in prior work, but in a different field and with a different outcome. With head-mounted eye-tracking, headset slip is known to degrade data quality. Even minor movements like speech or facial expressions can cause gaze tracking errors of 0.8–3.1° (Niehorster et al. 2020). The equivalent quantity for a VR HMD has not, to our knowledge, been measured, although it has been identified as a limitation of VR-based cervical assessment (Santos-Paz et al. 2022). The headset design, weight and fit have been found to affect the reliability of such assessments more than the tracking technology itself does (Santos-Paz et al. 2026), but a true measurement of the impact of these factors have not been published.

This work pursues two parallel lines of enquiry: whether the accuracy reported under rigid laboratory mounting persists when the same headset is worn by a freely moving individual, and how much the headset displaces relative to the head during use.

The contribution is a separate experimental characterization of both error sources in worn HMDs, using optical motion capture as an external reference.

## Methods

The investigation consisted of two experiments. The first compared the headset's reported pose against the optical ground truth in order to quantify the tracking error of the system; the second tracked the headset and the participant separately in order to quantify the movement of the headset relative to the head. The same motion capture system was used in both experiments (OptiTrack, 12 cameras, 100 Hz). 3 volunteers took part and all participants gave written informed consent.

Experiment A compared the headset's own reported pose against the optical ground truth while a participant wore a Meta Quest 3S device while moving freely. The pose the headset reports refers to an internal virtual point, the center eye anchor, displaced from the markers on the shell by an unknown but constant rigid transform, so the two streams cannot be compared directly. This transform was recovered by solving the hand-eye calibration problem  $A \cdot X = X \cdot B$ , a standard robotics formulation for the fixed relationship between two rigidly connected frames whose motions are observed separately; we used the dual-quaternion solution (Daniilidis 1999), which in comparable HMD



Figure 1. Experimental setup for Experiment B. Retroreflective markers are attached to the headset shell and to a close-fitting cap worn by the participant, defining two rigid bodies tracked independently by the optical system.

work was found to converge more consistently than alternative formulations (Banaszczyk et al. 2024). Once the transform was known, the reported trajectory could be expressed in the frame of the optical system and the remaining divergence reported as absolute pose error (APE). We depart from Banaszczyk et al. (2024) in that the two systems were started independently; the offset in recording start times was recovered before pairing by correlating velocity magnitude in each stream, a quantity invariant to the reference frame (Mair et al. 2011).

Experiment B placed a second rigid body on a close-fitting cap worn by the participant. The cap was taken as a proxy for the skull, and the headset pose was expressed in the cap's own reference frame, in which head motion cancels. Slip was defined as the deviation from a baseline established at the beginning of each recording, with the participant stationary and the placement of the HMD verified by the operator. Measurements were carried out with two HMDs (HTC Vive Pro, Meta Quest 3) under conditions in which the participant rotated their head about all three axes, produced exaggerated facial expressions, and deliberately displaced the headset, at fitting levels ranging from tight to loose.

Both experiments were summarized with root-mean-square error and percentiles, computed for Experiment A on the APE (Banaszczyk et al. 2024); we report the median and 95th percentile to characterize typical and upper-range errors. All comparisons between the two error sources are made at the 95th percentile.

## Results

Over a 30 s recording, the reported pose diverged from optical ground truth by 3.15 mm RMSE in translation (p50 2.65, p95 5.35, maximum 10.71 mm) and  $0.45^\circ$  RMSE in rotation. This falls within the range previously reported for rigidly mounted headsets (Banaszczyk et al. 2024), providing early evidence that the accuracy measured under rigid mounting is retained when the headset is worn.

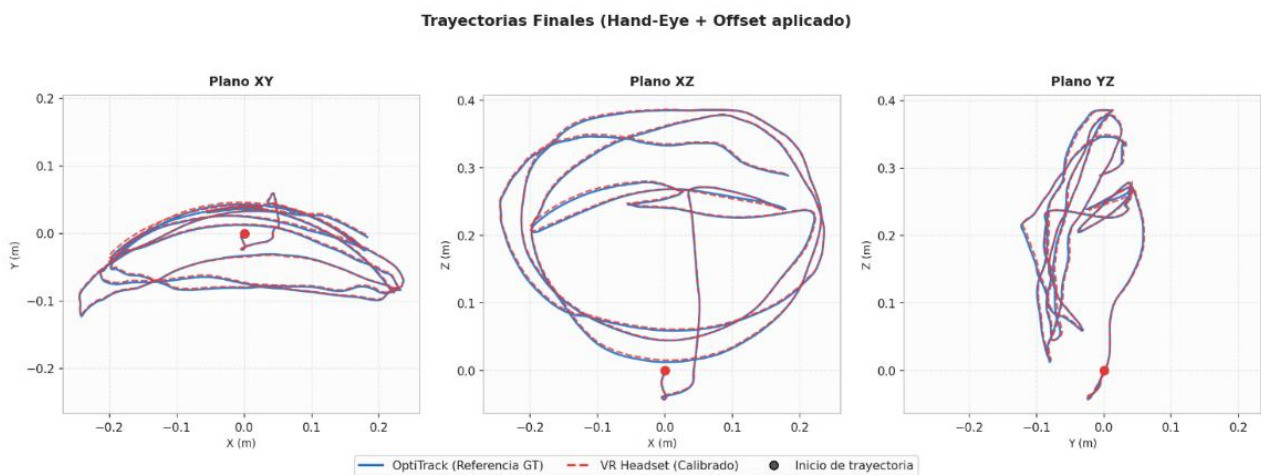


Figure 2. OptiTrack ground truth (blue) and the headset-reported trajectory (red) after hand-eye calibration and temporal alignment, projected onto the three coordinate planes.

As can be seen in Figure 2, fit mattered more than movement. With the participant sitting still, slip reached 0.13 mm at the 95th percentile, close to the measurement noise of the optical system. A carefully seated headset held to 0.78 mm even while the participant produced exaggerated facial expressions. Rotating the head about all three axes with a slightly loose fit

gave 1.29 mm on the HTC Vive Pro and 1.37 mm on the Meta Quest 3; a very loose fit reached 1.80 mm; and deliberately repositioning the headset mid-session, the worst case tested, reached 2.92 mm (all values at the 95th percentile).

How the headset was worn mattered more than how hard the participant moved: in two sessions in which the participant simply sat still, strap tightness alone changed slip by more than an order of magnitude (0.13 mm against 1.80 mm at the 95th percentile). Referred to the same physical point on the headset, device error exceeded slip in every condition tested, including deliberate repositioning, and was roughly seven times the slip of a properly seated headset (5.35 mm against 0.78 mm at the 95th percentile).

## Limitations and Future Work

The two errors come from separate recording sessions, because our current setup cannot track the cap, the headset and the headset's own recordings at the same time; we can therefore say how large each error is relative to the other, but not how the two combine. A further caveat is that the cap stands in for the skull: if cap and headset slid together the analysis would report no slip. Recordings lasted between 30 and 60 s, so we make no claim about how much error accumulates over a full therapy session, and the results rest on a small sample (3 participants), so the absolute magnitudes may not generalize. All these are being addressed: simultaneous capture of all three streams, separating the two errors within a single recording; markers fixed over the mastoid processes in place of the cap; longer recordings spanning a full therapy session; and a larger, more varied sample.

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